Manav Garg

Bachelors of Technology, Computer Science Engineering

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Experience

Vault22 - [*Remote*] Oct. 2025 – Present

Independent Consultant - AI/ML

- Led cost-optimization initiatives that reduced overall infrastructure and pipeline operating costs by 20–25% through workload restructuring and resource-efficient automation.
- Optimized and deployed AWS services to support scalable, self-hosted model deployments with improved performance reliability and reduced compute overhead.
- Performed extensive data cleaning, resolved ETL pipeline failures, and restructured data lakes to improve data integrity, consistency, and accessibility across teams.
- Designed and built analytical workflows and KPI dashboards using Amazon QuickSight, enabling accurate metric tracking and improving decision-making for business leadership and investors.
- Designed and build custom end tot end admin portals for the company to provide critical analysis about inner workings of the organization.
- Played a role in optimizing mongodb storage collections and indexes as well as optimizing backend pipelines to mitigate errors.

IHX Pvt. Ltd - A Perfios Company - [*Bangalore, Onsite*] Jan. 2025 – March 2026

Associate Data Scientist

- Led research-driven initiatives in tariff digitization, automating data extraction and standardization for large-scale financial datasets with measurable gains in accuracy and efficiency.
- Designed and deployed modular Python libraries for internal research and production pipelines, enabling reproducible ML experimentation across teams.
- Conducted applied research on multimodal architectures, integrating RAG + LLM frameworks with YOLO, Donut, and custom encoder–decoder models for complex document understanding.
- Excel in prompt engineering — systematically reduced hallucinations by over 70% and achieved consistent structured outputs even from non–fine-tuned, general-purpose LLMs.
- Pioneered optimized preprocessing and postprocessing pipelines, cutting data processing time from 24–36 hours to under 5 hours while improving dataset quality for downstream tasks.
- Developed scalable, asynchronous ML deployment systems to support continuous experimentation and real-time inference under production constraints.

Intel Corporation - [*Chennai, Onsite*] May. 2024 – Aug. 2024

Intel Unnati Industrial Trainee

- Classifying content within the Contract clauses.
- Determining deviations from Templates and highlight them.
- Provide accurate metrics for analysis.
- Utilized event driven pipelines and vision language models, to extract information with pre processing and post processing steps to filter and clean the data.
- Utilized semantic rich embeddings to extract useful information which then was used to detect fraud, reading between the fine print and to validate a contract.

Tisac pvt. ltd. - [*Remote*] May. 2024 – June. 2024

Intern, Full Stack Developer & ML Engineer

- Configured and administered cron jobs for streamlined task automation.
- Executed custom domain configurations for all API endpoints & managed reverse proxy setups to improve security and performance
- Deployed Jupyter Docker Lab environments for enterprise-grade data science applications
- Incorporated Firebase Authentication for secure and reliable user authentication
- Developed an advanced Docker API service for automating the creation of Jupyter Docker containers for enterprise model training
- Designed custom scripts to automatically assign secure TLS/SSL HTTPS subdomains to each container
- Ensured secure and efficient data storage solutions using MongoDB

Hogarth India – A WPP Company - [*Gurugram, Onsite*] Aug. 2023 – Oct. 2023

Intern, Full Stack Developer

- Part of **StudioX – The Coca Cola Company** project.

- Involved in developing a Full Stack Application for ICC World Cup 2023 Thums Up campaign to successfully log live votes, and Code Redemption as well live Polling.
- Developed and optimized Frontend for streamlined and production ready environment
- Developed and tested backend APIs developed using C# and .NET framework.

Wipro - [*Chennai, Onsite*] Aug. 2023 – Nov. 2023

Intern, Android Developer & Machine Learning

Contributed to the development of an android application utilizing machine learning for monitoring objects within the pathway obstructing the passage for their rovers and machines to move on. Part of the Wipro Infrastructure Engineering Department.

Education

Vellore Institute Of Technology, Chennai June 2025

B.Tech. Computer Science Engineering *Current CGPA: 8.51/10.0*

Specializing in: Artificial Intelligence and Machine Learning

Coursework

Courses: Object-Oriented Programming, Data Structures & Algorithms, Machine Learning, Artificial Intelligence, XAI, Linear Algebra, Calculus, Physics, Probability & Statistics

Skills

Languages: C/C++, Python, Java, JavaScript/TypeScript, HTML5/CSS3, Dart, SQL, Bash, LaTeX

Tools: Git/GitHub, Docker, Unix, PyCharm, Visual Studio Code, Vim, Adobe Photoshop/Illustrator, MongoDB, NoSQL Platforms, Vercel, Figma

Frameworks: ReactJS/NextJS/Express, Django/Flask/FastAPI, Node.js, WordPress, Material-UI

Libraries: Pandas, NumPy, Matplotlib, PyTorch, TensorFlow, OpenCV

Area of Expertise: ML-OPS, DevOPS, Full Stack Engineering, Data Analytics

Projects

LuminSkin: Skin Cancer Classification using Vision Transformers | *Python, PyTorch, ML/AI* Oct-Nov. 2024

- Achieving state-of-the-art performance in skin cancer classification, this project advances the capabilities of medical imaging analysis, providing enhanced tools for early detection and treatment. The approach leverages evaluation metrics such as classification accuracy to validate its effectiveness compared to existing methods. Additionally, this study compares the performance of two Vision Transformer models: the standard ViT and the DeepViT.
- LinkedIn · Github

Satellite Image Analysis using Vision Transformers | *Python, PyTorch, Machine Learning/AI* April-May. 2024

- Achieving SOTA in land use classification. This project contributes to advancing satellite image analysis and enhances tools for environmental monitoring, urban planning, and land management. This uses evaluation technique based on metrics like classification accuracy that proves the efficacy of the suggested approach with reference to existent methods. This project also compares two Vision Transformer models, the base ViT and the CCT (Compact Convolutional Transformer) model.
- LinkedIn · Github

KeyKeg - Password Wizard | *Python, Scikit-Learn, Machine Learning, NextJS, Web App* April. 2024

- Developed a Secure, Strong, and Smart, ultimate password management app that securely stores your passwords, generates strong and unique passwords, and utilizes machine learning approach (Random Forest Algorithm) to analyze and classify password strength.
- LinkedIn · Github

Fixed Point Adherence – Wipro | *Python, TensorFlow, Machine Learning, Flutter, Android Dev* July-Sept. 2023

- Developed an App for Wipro with Machine Vision capabilities to distinguish items on a pathway so that they can be removed and status can be logged.
- LinkedIn · Github

NSCHR, A Web HMIS Portal | *Express, Embedded JS, MongoDB, CSS3, Git, Unix Shell, VS Code* Sept. 2021

- A Web app built using various technologies to provide fast, accurate information about a patient right in front of Medical Practitioner in just a matter of few clicks!
- Github